

Aufg. 1

(a)

$$G_{s11} = \frac{4}{1+s \cdot 2} + \frac{-\frac{2}{3} \cdot 3}{1+s \cdot 2} = \frac{2}{1+s \cdot 2} \quad ; \quad G_{s12} = \frac{-\frac{2}{3} \cdot \frac{3}{2}}{1+s \cdot 2}$$

$$G_{s22} = \frac{3/2}{1+s \cdot 2} \quad ; \quad G_{s21} = \frac{3}{1+s \cdot 2}$$

$$\Rightarrow \underline{G}_s = \begin{bmatrix} \frac{2}{1+s \cdot 2} & \frac{-1}{1+s \cdot 2} \\ \frac{3}{1+s \cdot 2} & \frac{3/2}{1+s \cdot 2} \end{bmatrix} \checkmark$$

(b)

$$\underline{G}_K(s) = \frac{1}{\det \underline{G}_s} \begin{bmatrix} G_{s11}G_{s22} & -G_{s12}G_{s22} \\ -G_{s21}G_{s21} & G_{s11}G_{s21} \end{bmatrix}$$

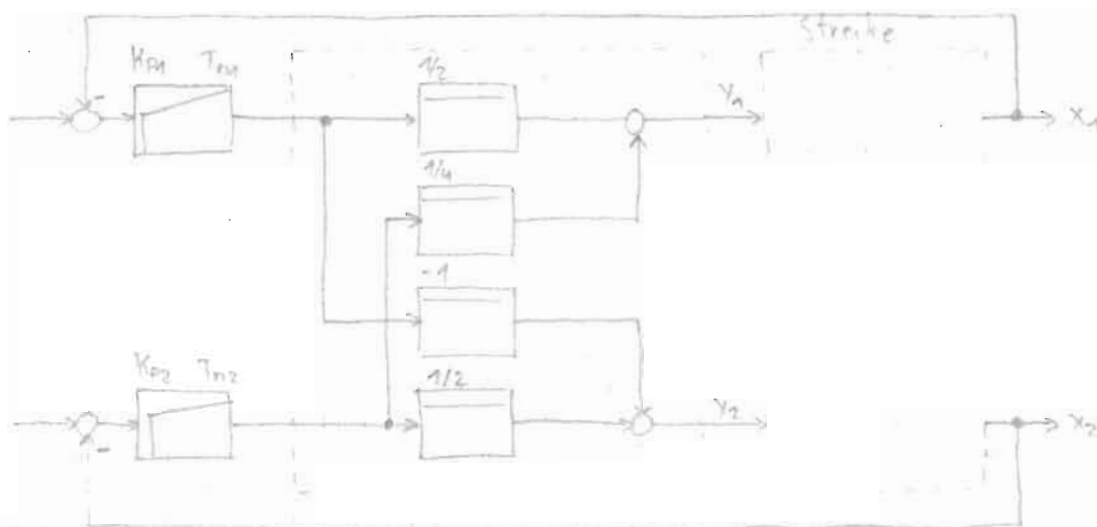
$$\det \underline{G}_s = \frac{6}{(1+s \cdot 2)^2}$$

$$\underline{G}_K(s) = \frac{(1+s \cdot 2)^2}{6} \begin{bmatrix} \frac{3}{(1+s \cdot 2)^2} & \frac{+3/2}{\dots} \\ \frac{-6}{\dots} & \frac{6}{\dots} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ -1 & 1/2 \end{bmatrix}$$

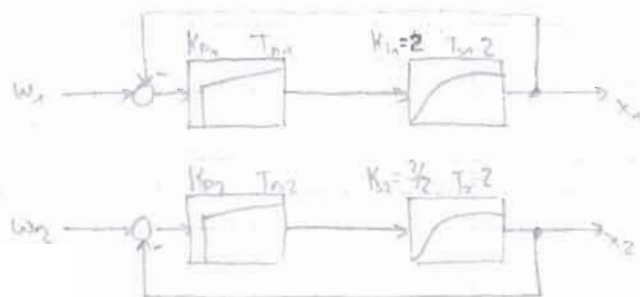
Probe:

$$\underline{G}_s \underline{G}_K = \begin{bmatrix} \frac{2}{1+2s} & 0 \\ 0 & \frac{3/2}{1+2s} \end{bmatrix}$$

(c)



||> (Skizze)



$$K_{p1} = \frac{T_{n1}}{K_{s1}} = \frac{2}{2} = 1 \quad \checkmark$$

$$T_{n1} = T_{s1} = 2$$

$$T_{n2} = T_{s2} = 2$$

$$K_{p2} = \frac{T_{n2}}{K_{s2}} = \frac{2 \cdot 2}{3} = \frac{4}{3}$$

(d)

$$G_0 = G_R \cdot G_s = K_p \frac{1+sT_n}{sT_n} \cdot \frac{K_s}{1+sT_s} = \frac{z_0}{N_0} \quad \Bigg| \quad G_w \stackrel{!}{=} \frac{1}{1+s}$$

$$G_w = \frac{z_0}{z_0 + N_0} = \frac{K_p K_s (1+sT_n)}{K_p K_s (1+sT_n) + (1+sT_s) s T_n} \stackrel{\text{mit}}{=} \frac{1}{1+s \frac{T_n}{K_p K_s}} \Rightarrow \frac{T_n}{K_s K_p} \stackrel{!}{=} 1 \Rightarrow K_p = \frac{T_n}{K_s} \quad \checkmark$$



(b) $N(u) = u + \frac{1}{2} \sin u \stackrel{!}{=} v$

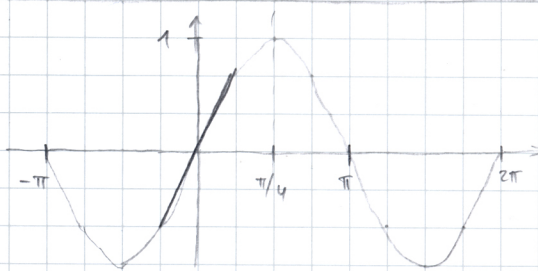
$K \stackrel{!}{\geq} 1$ (Schwankung d. Verstärkung)

Linearisieren:

$$\tilde{v} = v_R + \left. \frac{\partial v}{\partial u} \right|_{u_R} (u(t) - u_R)$$

$$K = \left. \frac{\partial v}{\partial u} \right|_{u_R} = 1 + \frac{1}{2} \cos u_R \stackrel{!}{\geq} 1 \Rightarrow \cos u_R \stackrel{!}{\geq} 0 \Rightarrow |u_R| \stackrel{!}{\leq} \frac{\pi}{2}$$

oder $-\frac{\pi}{2} \leq u_R \leq \frac{\pi}{2}$



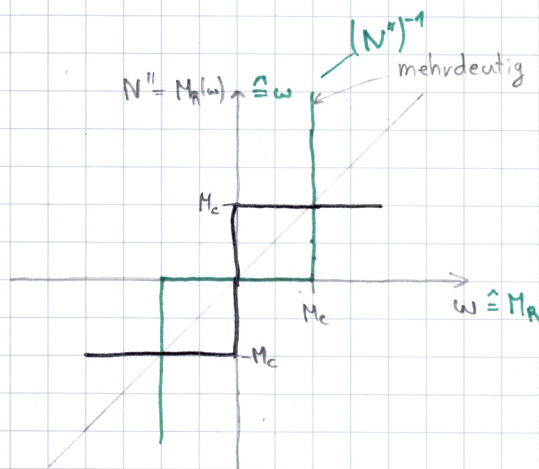
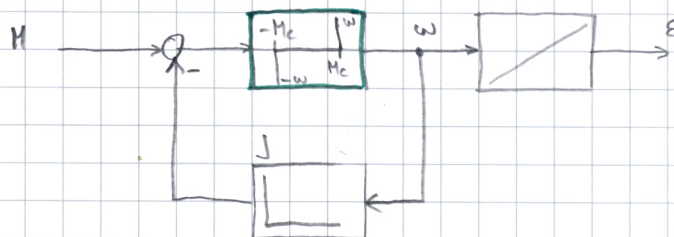
(c) 2. Standardform

$$G_c'' = G_c' = \frac{1}{s}$$

$$G_d'' = \frac{1}{s \cdot J} \Rightarrow G_d' = \frac{1}{G_d''} = \frac{s \cdot J}{1} \stackrel{!}{=} T_D$$

$$N'' = M_c \cdot \operatorname{sgn}(\omega)$$

$$\Rightarrow N' = N''^{-1}$$



mathematische Beschreibung

$$N' = [N'']^{-1} = \omega = \begin{cases} 0, & -M_c \leq M_R \leq M_c \\ -\omega, & M_R = -M_c \\ \omega, & M_R = M_c \end{cases}$$

$$- \omega, \quad M_R = -M_c$$

$$\omega, \quad M_R = M_c$$

$$- \text{undef.} \quad \text{sonst}$$